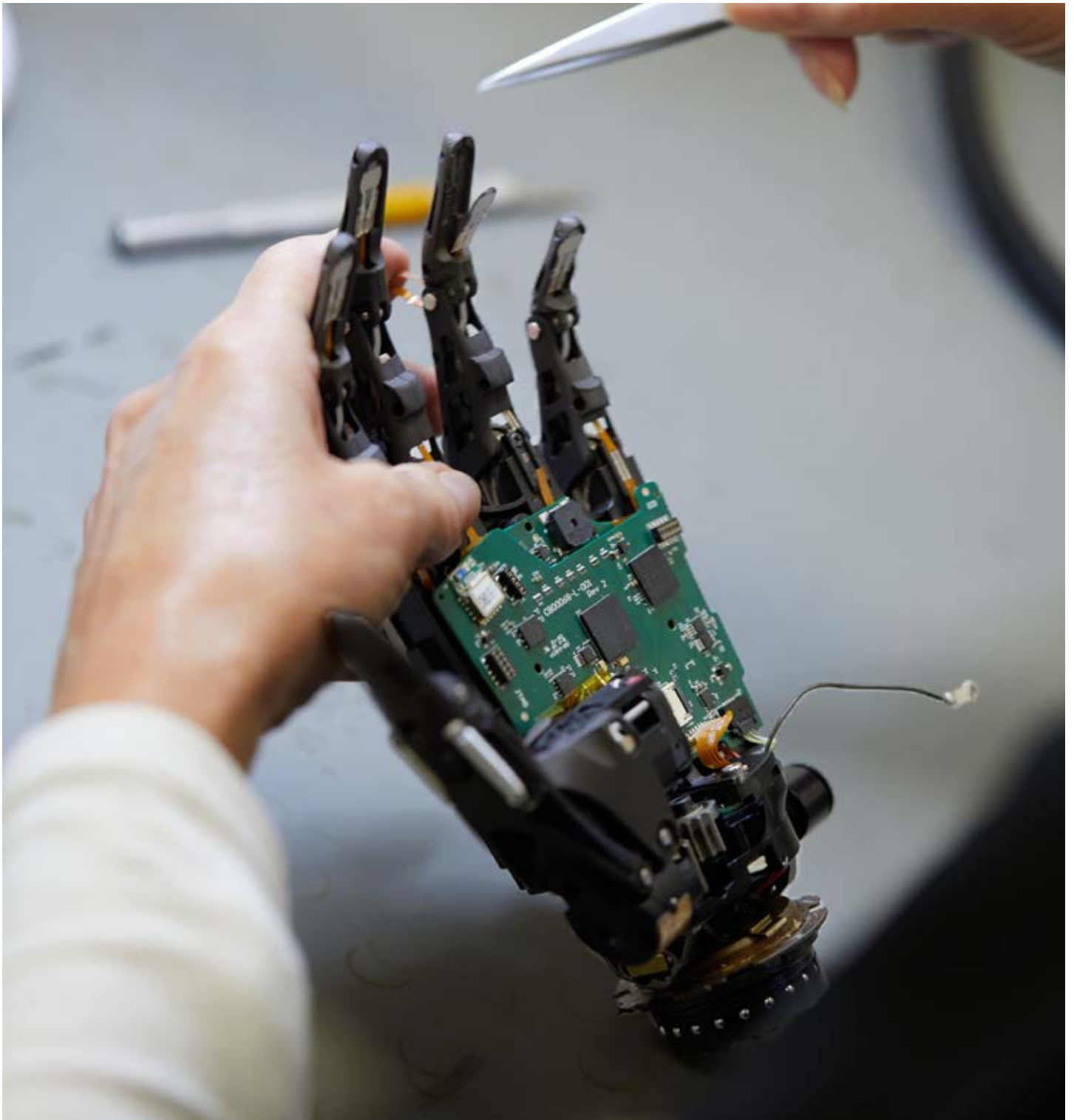




COVVI ROBOTICS

Sales Brochure



WHO ARE WE?

Our robotic device has been derived from the world-leading COVVI Prosthetic Hand

With over 75 years of combined experience in prosthetic design; COVVI's dedicated team has spent the last five years creating a world-leading, multi-articulated hand that is both highly functional and accessible.

We are now introducing COVVI Robotics, bringing the ground-breaking COVVI Hand, a state-of-the-art multi-articulated hand with the ability to complete complex manipulation tasks, to the robotics sector.

Our clients are at the heart of everything we do and we're passionate about delivering a first-class customer experience.

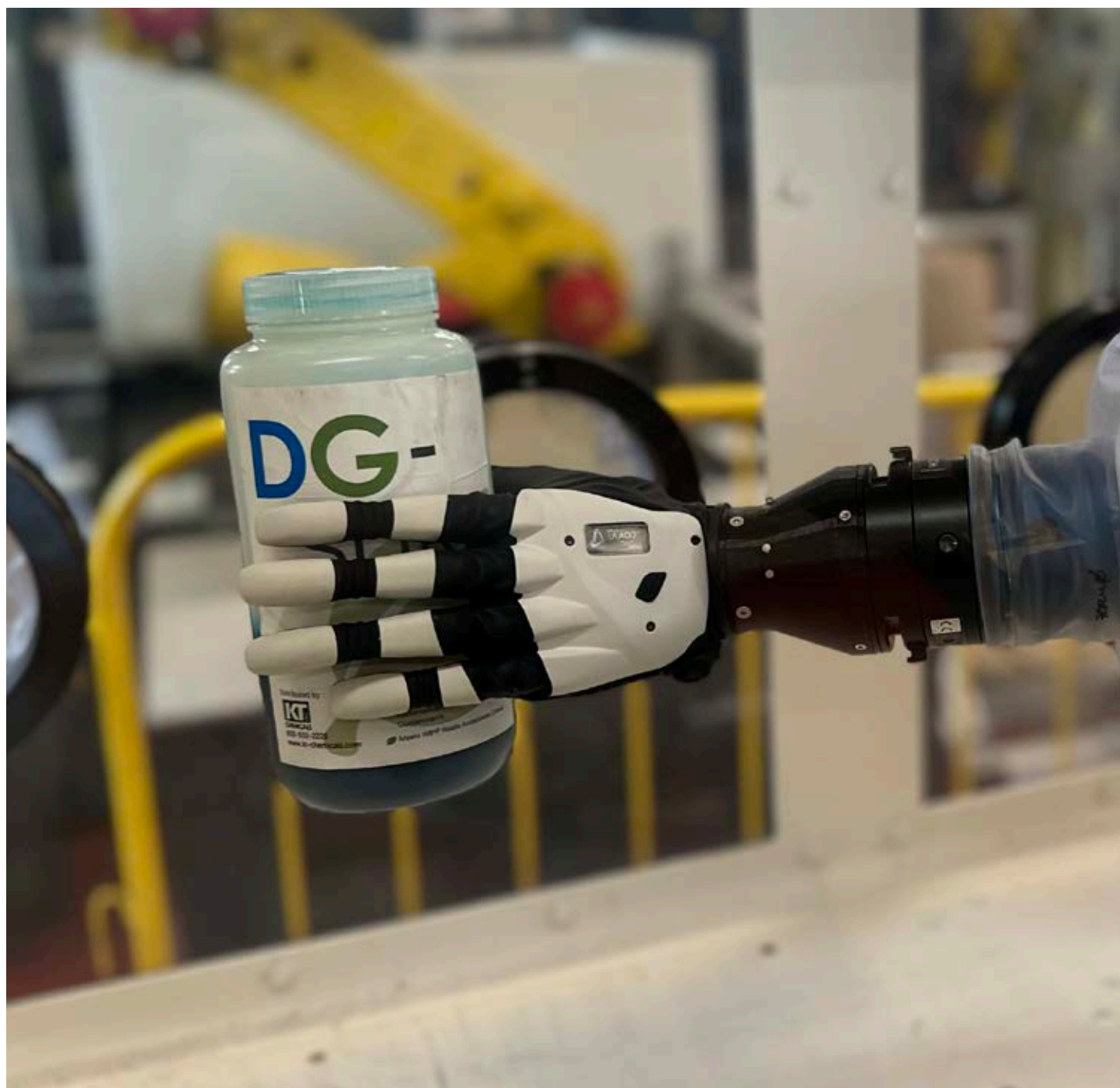
DEVICE OVERVIEW

The COVVI Robotic Hand is a five-finger end effector with six degrees of freedom. Adapted from the COVVI Prosthetic Hand, the COVVI Robotic Hand mimics human hand functionality, capable of completing a wide range of actions using 14 pre-programmed grips and individual digit control.

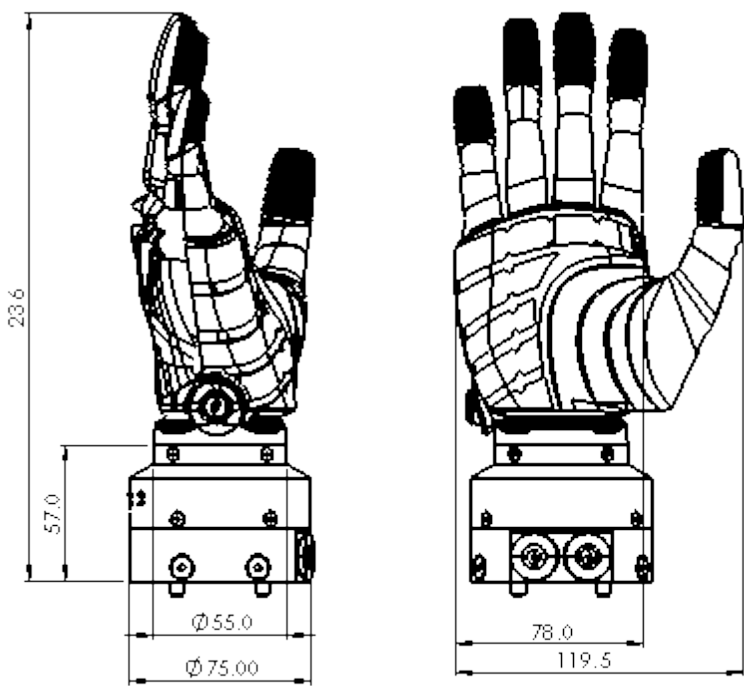
The COVVI Robotic Hand is powered through a 24v supply and supports ethernet communication and control via the ROS 2 Software Interface.

Capable of individual digit drive, as well as providing finger flexion and thumb rotation and flexion, the hand can securely grip a wide range of objects with a maximum payload 30kg and grip strength of up to 90N. With integrated force sensing in the fingers and specialised anti slip glove fingertips.

The COVVI Robotic Hand is capable of replacing a human hand in a large range of robotic operational environments.



TECHNICAL SPECIFICATIONS



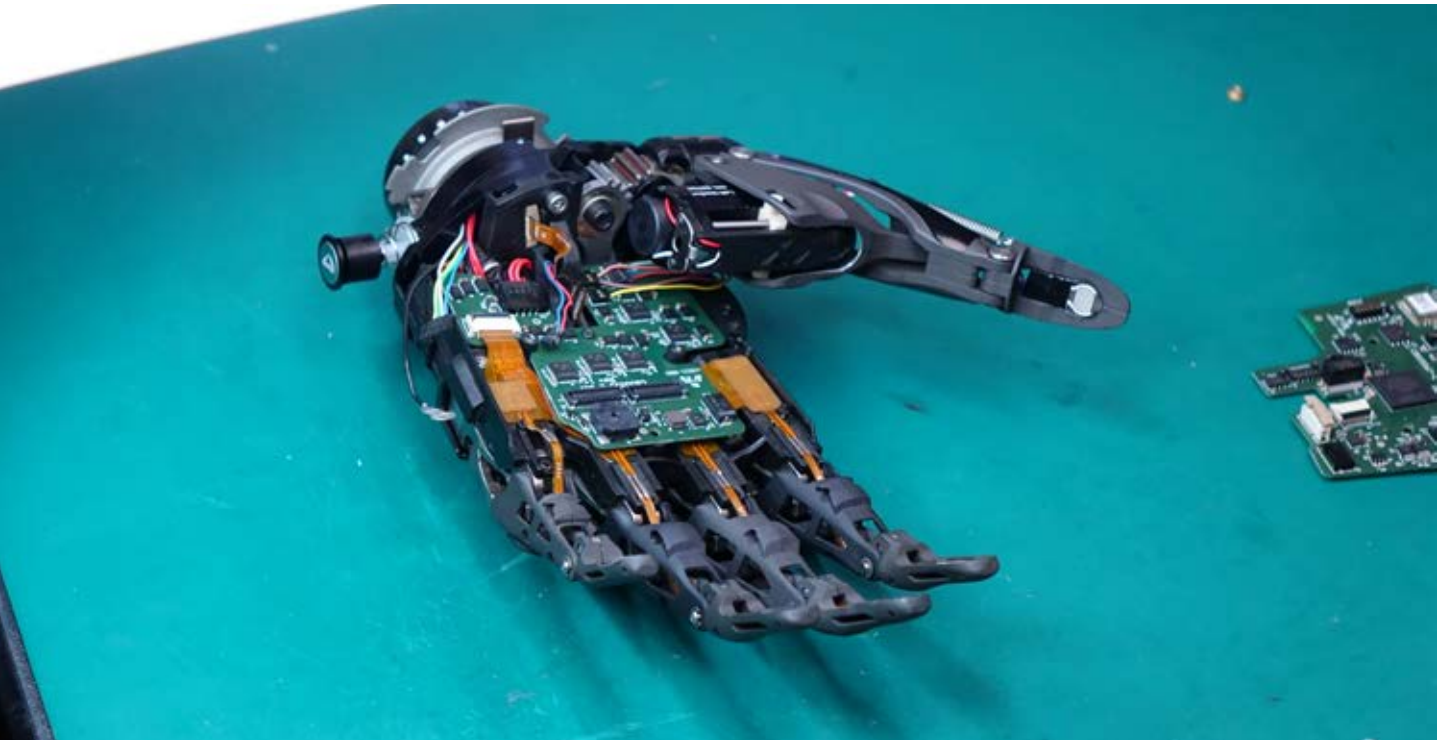
Dimensions	Height	Width	Weight
COVVI Robotic Hand	179mm	75mm	650 grams (+/- 5g)
Wrist Adapter	57mm	75mm	350 grams (+/- 5g)
Total System	236mm*	75mm	1 kg (+/- 10g)

Specifications	Value
Noise	< 72dB
Voltage Input	24V
Communication Interface	Ethernet
Control	ROS 2 Driver
IP Rating	IP44

*Base of adapter to top of fingertip

PERFORMANCE INFORMATION

Specifications	Value
Degrees of Freedom	6 - Individual digit flexion, thumb rotation and flexion
Maximum Hand Load Limit	30 kg / 66 lbs across the knuckles
Maximum Hand Load Limit	10 kg / 22 lbs in power grip
Maximum Digit Load Limit	16 kg / 35 lbs on each finger
Full Open to Full Close	0.6 seconds
Tripod Grip	0.4 seconds
Tripod Grip Force	35N
Power Grip Force	90N
Key Grip Force	22N
Control	ROS 2 Driver



MECHANICAL STRUCTURE



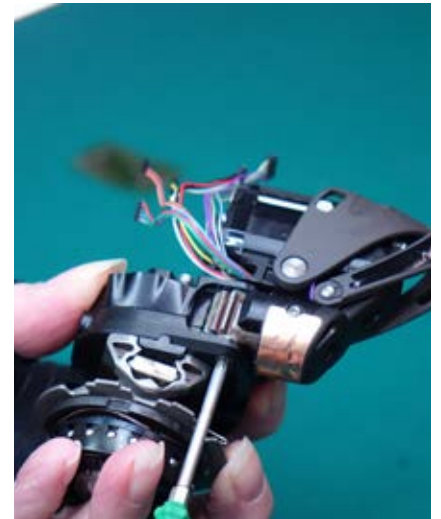
Improvement

We endeavour to continuously improve our designs and listen to our users. Based on partner feedback, the design of the COVVI Robotic Hand has been focused on durability and function.



Premium Materials

Based on our past designs and knowledge, we recognise weaknesses and have chosen to reinforce components and use durable materials.



Modular Design

There are approximately 450 components in our hand, each designed in-house by our experienced engineering team.



KEY FEATURES

- **Low Energy Consumption (24V)**
- **Individual Digit Control**
Both thumb flexion and rotation can be separately controlled.
- **Pre-Programmed Grips**
14 grips modelled after the most used human grip patterns.
- **Non-Back Driveable Digits**
- **Digits are compliant in the close direction**
- **Compact Design**
Created with consideration of the average human hand dimensions, our design results in a compact and lightweight robotic hand package.
- **Supports Internal and External Wiring**
Both power and ethernet connections can be connected internally through the robotic arm or externally through the wrist adapter connectors.
- **ISO 9409-1 Compatible Interface**
Ensures the exchangeability and keeps the orientation of the hand-mounted end effectors.
- **Protective Glove**
The protective glove helps protect the COVVI Robotic Hand from liquid and debris ingress.
- **Anti-Slip Fingertip Design**
The silicone glove fingertips have been designed to provide secure grip throughout a series of grip patterns.
- **Integrated Sensing**
Force Sensitive Resistors integrated into the fingertips of the COVVI Robotic Hand provide feedback to the user.
- **Built To Last**
The device will have an estimated lifetime of 500,000 cycles, with recommended services at every 100,000 cycle interval for maintenance.



ROBOTICS INNOVATIONS

In order to create a device that is tailored to the robotics market, we have adapted several aspects of our original prosthetics hand to better suite the unique tasks that our robotic hand with undertake. These include:

- **Thumb Flexion Assembly**
New fixed thumb pivot gear, with the thumb rocking prosthetic mechanism removed.
- **Thumb Flexion and Rotation Assembly**
New thumb rotation lower plate that allows wrist adaptation to be screwed in from underneath.
- **Wrist Adapter Assembly**
Houses the power conversion and ECI board as well as an ISO 9409-1 Compatible Mechanical Interface to allow for mounting on co-bots.
- **Functional Assembly**
Combination of the new assemblies into a functional part.
- **Adapted Glove**
Combination of the new assemblies into a functional gloved part.



GLOVE DESIGN

The protective COVVI Glove is an IP44-rated replaceable component of the COVVI Robotic Hand, providing protection to the internal components and anti-slip fingertips ensuring secure grip on objects.

Made from silicone with a soft-touch coating, the gloves boast thicker integrated fingertip pads to improve the life of the glove.

At every stage of design, we've thought about how the environment and materials that are handled by a robotics, so we ensured that our glove was strong and robust enough to handle these tasks at a consistent level. The newest iteration of our glove design features:

- **Improved longevity and durability**
- **Knuckle bumpers for additional protection**
- **Harder fingertips with optimised geometry**
- **Thicker palmar region**





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Phone: +44 (0)20 3949 9500
Email: customerservice@covvi.com
Website: www.covvi-robotics.com



COVVI Ltd.,
Unit 4 (Direct House),
Quayside Business Park,
Leeds, LS10 1DJ